# Fixed Unmanned Arial Vehicle Launch System
## Project Proposal

MREN 403

Prepared by: James Pankiw, Jamison McCulloch-Cop, Darcy Ryan, and Ahrvin Raveendran

Supervisors: Jonathan Gammell and Melissa Greeff

October 9, 2026

# Cover letter

To Jonathan Gammell and Melissa Greeff,

MREN 403 Group 10 — James Pankiw, Jamison McCulloch-Cop, Darcy Ryan, and Ahrvin Raveendran — submits this proposal for a gated launch and one pitch axis for the MakeFlyEasy Fighter Hand Throw from a Zodiac Nautic PRO 7. We are excited for this project as it gives us a real engineering problem with clear value for search-and-rescue crews and a team well suited with experience to solve it.

The MakeFlyEasy Fighter hand-thrown UAV is a capable search-and-rescue aircraft, but only can be launched from land. This proposal is a rail launcher for the Zodiac Nautic PRO 7 to extends its value of search-and-rescue to the waters. The design rests on three core ideas. First is the two meter rail, winch and latch, which accelerates the aircraft to 15 m/s in about 0.27 s, with consideration to protect the sensor payload, wings, and propellers. Second, there is a pitch gate that uses two cosine forecast of hull motion to release the aircraft only when its launch window is within 0-10°, and if no safe window exists, it will terminate the launch. Third, a single pitch axis holds the rail steady through the launch window, validating a safe launch is available.

The work has been scoped to what can be validated and this term we will build a scaled prototype within the budget of $500 and test each subsystem on the bench against a defined criteria. The criteria will include the pitch gate, pitch stabilization, carriage, structure and a scaled-drone launch. The objectives are outlined in Section 6, and the weekly plan and deliverables are in Section 7 and Section 8. A boat trial will follow only if the scaled-prototype is validated on time and supervisors approve to move further.

Our team brings experience in mechanical design, controls and software, and marine rescue operations alongside the Coast Guard.

Thank you for reviewing our proposal. We welcome your feedback as we look forward to delivering the validated prototype.

Sincerely,

James Pankiw 

Jamison McCulloch-Cop 

Darcy Ryan 

Ahrvin Raveendran 

# Contents


Cover letter ..... 2
1. Executive summary ..... 4
2. Purpose ..... 4
3. Project scope ..... 4
4. Review of state of the art ..... 5
4.1 Sears, wave mapping ..... 5
4.2 Insitu, launch carriage ..... 5
5. Technical background ..... 6
5.1 Rail ..... 6
5.2 Brake ..... 7
5.3 Pitch of the hull ..... 7
5.4 Forecast ..... 7
5.5 Wind ..... 8
5.6 Assumptions ..... 8
6. Objectives ..... 9
Objective 1: Launcher design ..... 9
Objective 2: Pitch stabilization ..... 9
Objective 3: Control and launch prediction ..... 9
7. Deliverables ..... 9
8. Schedule of work ..... 10
9. Budget ..... 11
10. Validation ..... 12
11. Team qualifications ..... 12
References ..... 14

## 1. Executive summary

The non-VTOL MakeFlyEasy Fighter cannot be launched safely by hand from a Zodiac Nautic PRO 7. This project designs a rail launcher that releases the aircraft only when a short pitch forecast says elevation will stay inside 0–10°, on a heading into the waves, with the propeller already at full throttle and a person committing the latch. A brake at the end of the stroke opens the gripper and takes the forces on the fuselage, so the payload, wings, and props stay protected. One pitch axis holds the rail in the launch window. The full design mounts modularly on the PRO 7 without modifying the hull. This term builds a scaled-down prototype inside the $500 course fund. Pitch gate, pitch stabilization, carriage, and structure will be validated on the bench with scaled forces and a scaled drone. A boat trial is conditional on successful and timely validation of the prototype.

## 2. Purpose

The purpose of this project is to design and build a fixed-wing UAV launching system for the Zodiac Nautic PRO 7, a 6.90 m (about 23 ft) rigid-hull inflatable in the 20–25-foot class used by coast guard crews. The UAV is the MakeFlyEasy Fighter Hand Throw (Big Bird), a search-and-rescue aircraft for use over water. Launching from the PRO 7 widens the situations in which the coast guard can use it. The aircraft can currently be launched only from land. Angle, speed, and stability make a launch from the vessel a problem in both hardware and control. The purpose is a launch that is safe and repeatable from the PRO 7 in open water. Accelerating the aircraft through the fuselage alone may pull the wings off, so the carriage may also have to bear on the wings.

Stakeholders for this project include the Coast Guard crew who must launch safely without damage to the boat or risk to passengers, the search-and-rescue users of the Fighter, and the course supervisors who require a bench-validated prototype inside the $500 fund before any boat trial.

## 3. Project scope

The full design is a launcher for the MakeFlyEasy Fighter Hand Throw on the Zodiac Nautic PRO 7. The work covers launching hardware and control software. A measured motion window is the no-fire authority: if no safe window is found, the system does not launch. One pitch axis holds a launch elevation inside the decided window (±5° about the set angle, within the 0–10° band). Roll and yaw are added only if a later log shows that pitch is not enough. The boat heads into the waves, and launch is allowed only when the apparent wind stays inside a ±30° band of that heading. That ±30° figure is a proposed operating limit, not a measurement. On the full system an anemometer supplies the wind check; the Coast Guard boat is not assumed to already log gusts. Assumed seas for the pitch model match the technical background: crest-to-trough heights of 0.30–0.50 m and crest-to-crest wavelengths from about 1.6 m (1 s chop) to about 25 m (4 s wind wave).

The release concept is a 2 m rail, winch, and latch, with an assumed exit of 15 m/s in 0.27 s. The brake opens the gripper. A trebuchet is ruled out. The propeller must be at full throttle before the latch opens, and the hold-back is sized to the measured static thrust; the Fighter page gives no thrust. The launcher must protect the sensor payload, propellers, wings, flaps, and tail: nothing may break or bend, and the payload must not be scratched or obstructed. The rail structure uses a modular mount on the PRO 7 that attaches and removes without modifying the

hull or endangering passengers. It carries the force along the rail (under 390 N at 7 kg), the full-throttle static thrust at the latch, and the brake load at the end of the stroke, and it fits inside the PRO 7's 5.71 m by 1.39 m inside deck. Structural sizing follows the proposal; there is no structural model yet.

This term's build is a scaled-down prototype of that physical system, kept inside the $500 capstone budget: the pitch gate, a short pitch axis, a hold-back static pull, and a short rail, brake, and gripper on the bench. It is not a full-speed model of the 2 m boat rail. The anemometer and the boat mount are omitted from the prototype buy. Wind-band logic is tested with recorded direction instead.

Out of scope are a two-axis gimbal and launch-in-any-sea as the success criterion, a learned model that arms the latch, a full at-sea trial as the fall deliverable, and operation in snow, hail, or heavy rain.

# 4. Review of state of the art

## 4.1 Sears, wave mapping

Sears, Cooper, and Marshall [1], Mapping waves with an uncrewed surface vessel via Gaussian process regression, mapping the wave field from an Otter USV on Lake Ontario. Station-keeping tilt spectra peaked at around 1 Hz. Thomas Sears advised that thesis chapters 4 and 6 apply, and that a prediction of about 0.5 s from a few seconds of prior data is feasible.

This launch takes the time-only form of the spectral mixture in Sears [2]: two cosines fit to pitch at one point, over the rail time of about 0.27 s. The spatial map, the product-of-sums kernel, and a rotated wave heading belong to mapping the lake and are not this gate. A smooth kernel returns the mean past the samples, so that form is not used for the extrapolation.

Sears also has a simulation that builds the waves with a longitudinal model and drives the boat mechanics from parameters. It is not a published paper. His simulation is the closer example of driving hull motion from wave and boat parameters. This proposal uses a simpler rigid-hull slope, with the PRO 7 length and beam. The gate does not run his boat-mechanics model. Further simulation will be done once the group receives that code.

## 4.2 Insitu, launch carriage

Insitu, U.S. Patent 7,128,294, Methods and apparatuses for launching unmanned aircraft, including methods and apparatuses for launching aircraft with a wedge action, is the ScanEagle SuperWedge catapult [3]. A brake at the end of the stroke is separate from the acceleration runway. The gripper's arms are linked to the carriage, so they meet the brake with it. Figs. 6A and 6B show the aircraft on the carriage, and the arms swung outward.

The group will use that braking mechanism to release the gripper arms, and to take the forces on the fuselage. When the carriage stops, the arms still have forward momentum and swing clear. That is the idea carried into this launch. It is how the group will address the problems raised in the problem definition. The patent is not a pitch gate, and the drawing gives no spring force, angle, or load to copy.

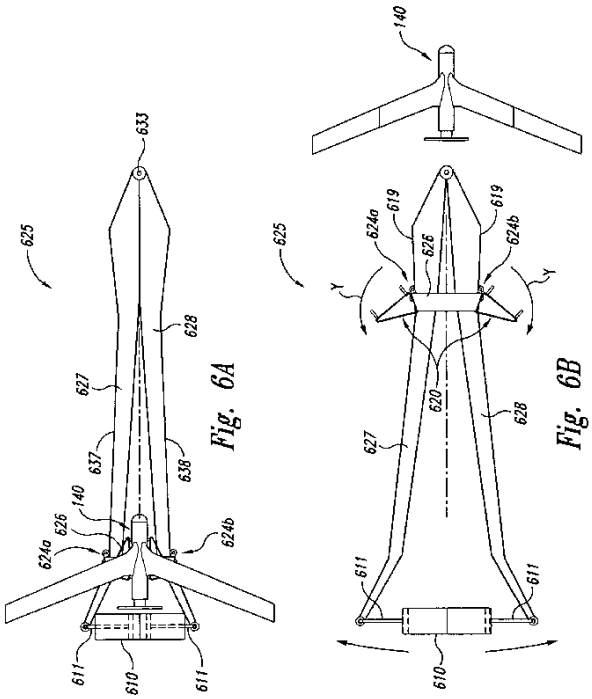



Figure 1: Figures 6A and 6B from inisitu patent showing carriage release and braking mechanism [3]

# 5. Technical background

## 5.1 Rail

The launcher is a straight 2 m rail on the Zodiac Nautic PRO 7. The boat's inside length is 5.71 m. The 2 m stroke is the runway, the carriage, the brake, and the gripper use [5]. A standing throw on the moving boat is unsafe for the crew, and the hull changes the exit attitude during the throw.

A trebuchet, or any throwing arm design, was set aside. The arm releases the aircraft at one angle, so a pitch axis cannot hold elevation through the launch. The arm and its counterweight need a swing circle beside and above the deck. The PRO 7's overall beam is 2.54 m, and the crew is already on that deck. The release is a sling rather than a carriage meeting a brake, so the force into the fuselage arrives as a jerk. Accelerating the aircraft through the fuselage alone may already pull the wings off, and a jerk makes that worse. The rail keeps the force along the fuselage for the whole stroke.

The carriage is assumed to start from rest and leave the 2 m rail at 15 m/s. The Fighter page gives no throw speed. Constant acceleration then fixes the acceleration and the time on the rail:

$$a = \frac{v^2}{2s} = \frac{15^2}{2 * 2.0} = 56\frac{m}{s^2} \approx 5.7g, t = \frac{2s}{v} = 0.27s$$

At a takeoff weight under 7 kg, the force along the rail is under $7 \times 56 \approx 390$ N. The kinetic energy is under $\frac{1}{2} \times 7 \times 15^2 \approx 800$ J. The 2 m length is the acceleration stroke. Whether the brake sits beyond that stroke is still open. This term's build is the bench gate, not the boat rail.

## 5.2 Brake

The group will use a brake at the end of the stroke to release the gripper arms and to take the forces on the fuselage. The arms travel with the carriage. When the carriage stops, the arms still have forward momentum and swing clear. That is the release taken from Insitu, U.S. Patent 7,128,294, Figs. 6A and 6B [3]. How much of the launch force must also go into the wings is named in the problem statement.

The propeller is at full throttle before the latch opens. The hold-back has to resist that static thrust as well as the force along the rail. No thrust is published for the Fighter. The number comes from a static pull: a load cell in line with the latch, motors at full throttle, carriage not moving. The aircraft is not launched for this test.

## 5.3 Pitch of the hull

Once the boat heads into the waves, pitch is the motion the gate uses. The PRO 7's overall length is 6.90 m and its overall beam is 2.54 m. The pitch model uses Lb = 7 m, which is inside the sheet's ±3% tolerance on that length. Deep-water wavelength is $L = \frac{gT^2}{2\pi}$. Pitch amplitude of a rigid hull in head seas is:

$$\theta_{\max} = \frac{H}{L_b}\sin(\pi * \frac{L_b}{L})$$

Values H and T are assumed seas and are contained in Table 1 below. These are not experimental values and will be updated when new information becomes available.

Table 1: Wave parameters, pitch amplitudes, and rate-hold errors for different assumed sea states.

| **Assumed sea** | **H (m)** | **T (s)** | **L (m)** | **Pitch** | **Rate-hold error at 0.3 s** |
|---|---|---|---|---|---|
| Small wind wave | 0.30 | 3 | 14.05 | ±2.46° | 0.47° |
| Moderate wind wave | 0.50 | 4 | 24.98 | ±3.16° | 0.34° |
| Short chop | 0.50 | 1 | 1.56 | ±4.09° | 5.35° |

These heights do not set an actuator stroke. On the longer waves, a fraction of a second at a crest barely moves the pitch. On the 1 s chop, holding the crest angle is already wrong by about 5° before the aircraft leaves the rail. The gate has to use pitch rate as well as pitch angle.

## 5.4 Forecast

The gate predicts pitch with two cosines fit to the last 6 s of IMU data. This is the time-only form of the spectral mixture in Sears (2026, Chapter 4):

$$\theta(t) = \Sigma_{q=1}^{2} A_q \cos(2\pi f_q t + \phi_q)$$

Sears, Cooper, and Marshall [1] found that the tilt spectra of a small vessel on Lake Ontario peaked near 1 Hz, so a few cosines fit the motion. Sears (2026) showed that this form carries periodic motion past the last sample, where a smooth kernel returns to the mean. He advised that about 0.5 s of prediction from a few seconds of data is feasible. The rail time is 0.27s.

The control sequence is:

1. The operator gives a "launch when ready" input on the interface.
2. The motors run up to full throttle while the latch holds the carriage.
3. At every step, the gate predicts the next 0.27 s and checks the wind band.
4. The latch opens when the predicted elevation stays inside 0–10°.
5. If no release is allowed within 20 s, the launch is aborted and must be re-armed.

Figure 2 tests the gate on a synthetic sea that is harder than the model, with drifting wave heights, a 6s swell the fit does not model, and 0.3° of IMU noise over 1,201 release times. The left plot compares launch elevation for a fixed rail (black) and a stabilized rail (blue), with green regions indicating permitted releases using a 1.5° safety margin. The right plot compares false allows and missed launch windows. The forecast is about 1.3° off at 0.27 s, compared to 5.3° when holding the last rate. A fixed rail still allows unsafe launches 14% of the time, while the 1.5° margin reduces this to 2%. Stabilization also increases safe launch availability from 68% to 84%. These simulations will be further evaluated using Thomas Sears' wave simulations and the dimensions of the Zodiac Pro7 to model realistic vessel dynamics.

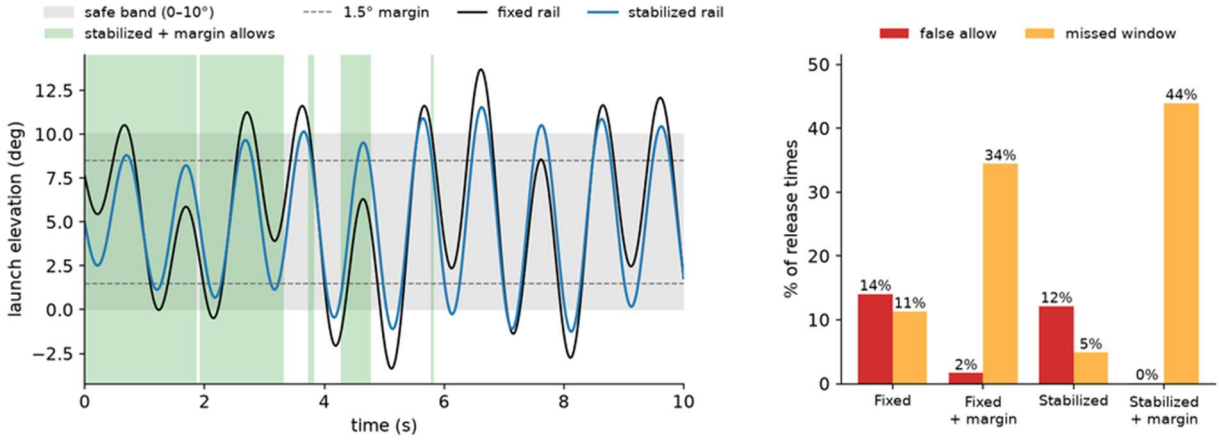



*Figure 2: Pitch-gate simulation on synthetic head seas for a 6.90 m hull.*

## 5.5 Wind

The boat heads into the waves. Launch is permitted only when the apparent wind stays inside ±30° of that heading. On the full system, an anemometer mounted with the rail supplies it; a 1 s average of direction is the check, logged on the same clock as the IMU. The scaled prototype omits the anemometer and tests that logic with recorded direction. Roll is a no-fire check that can be implemented pending prototype testing.

## 5.6 Assumptions

Exit speed, 15 m/s. Wind angle, ±30° from launch direction. Propeller thrust, unmeasured until the static pull. The pitch heights in the table are assumed seas. The stabilizer leaves 20% of slow motion in the simulation until the actuator and mount, as well as controller. The 1.5° margin is a starting value for the fall test.

# 6. Objectives

The overall goal is to design a launcher for the non-VTOL MakeFlyEasy Fighter on the Zodiac Nautic PRO 7. The system has three parts: a mechanical launcher that accelerates the UAV to flight speed, a pitch stabilization mechanism that holds the launch angle, and a controller that predicts boat pitch and permits launch only in a suitable window. The launch must not damage the UAV, the sensor payload, or the boat, and must remain safe for people on board.

## Objective 1: Launcher design

Design a mechanical and electrical launcher that reaches a minimum exit speed of 15 m/s over a 0.27 s rail time, with a clean brake and gripper release. The 15 m/s target is about 1.5 times a 10 m/s stall speed at the Fighter's takeoff-weight class, so the aircraft leaves above stall. The launcher must protect the aircraft and its sensor payload, hold the aircraft at defined points including the wings and tail, and clear the propellers, wings, flaps, and tail. The mounting must take the launch impulse without harming anyone on board. Light rain and salt spray protection are required on the full system; the fall prototype is a dry bench build. [4]

## Objective 2: Pitch stabilization

Design a mechanical and electrical system that holds launch elevation inside 0–10°, compensating boat pitch of about ±5° while the launch window is checked. In head seas, roll is treated as secondary and is a no-fire check rather than an active axis. The system must measure launcher pitch (angle and rate) with an IMU and mount securely on the vessel or the bench rig.

## Objective 3: Control and launch prediction

Develop a controller that predicts boat pitch over the launch window and permits a launch only if a suitable window exists. The prediction horizon must cover the 0.27 s rail time. A launch is allowed only when the predicted elevation stays inside the 0–10° band (within about ±5° of the set angle) for that horizon; otherwise it does not launch. If no release is allowed within 20 s of full throttle, the launch is aborted and must be re-armed. The predictor is a two-cosine fit on a few seconds of pitch, compared against holding the last pitch rate. On the full system, wind direction inside ±30° of the heading is an additional check; the scaled prototype tests that logic with recorded direction. The controller must run on low-cost onboard hardware with no network access.

# 7. Deliverables

Table 2: Table of deliverables breakdown

**Semester 1**

| Week | Deliverable | Weight |
|---|---|---|
| 9 | Engineering Portfolio I | 5% |
| 12 | Preliminary Design Review | 20% |

**Semester 2**

| Week | Deliverable | Weight |
|---|---|---|
| 4 | Engineering Portfolio II | 5% |
| 8 | Critical Design Review | 15% |

| | | |
|---|---|---|
| 11 | Prototype Demo and Technical Defense | 25% |
| 12 | Final Report | 10% |

## 8. Schedule of work

Fall focuses on the pitch gate, buying and building sensors and core components, and designing the rail and carriage. Winter adds waterproofing early, then pitch leveling (stabilization), and closes with structure, carriage, and launch validation before the prototype demo. No boat day is planned until the supervisors grant the hull.

Table 3: Schedule of work breakdown fall 2026 (semester 1).

| When | Work | Validation deadline |
|---|---|---|
| Week of 2026-10-09 | Project Proposal Submission | — |
| Through fall week 8 | Pitch gate software: IMU log, two-cosine forecast, Sears wave-model integration and simulation. Build and wire sensors and electronics on the bench (IMU, Pi, latch/hold-back sensing). | Successful testing of wave prediction model with Sears' wave simulation code. |
| Fall week 9 | Engineering Portfolio I (5%). | **Pitch gate:** wave model and simulation complete; forecast scored against a rate hold on the rocking rig. |
| Weeks 9–12 | Carriage gripper linkage designed and prototyped. Rail designed and built for the scaled prototype. Structure framed for scaled launch loads. Remaining mechanical components fabricated or 3D-printed. | Bench carriage and rail assembled enough to dry-cycle. |
| Fall week 12 | Preliminary Design Review (20%). Built sensors/components, rail, and carriage/gripper shown. Designs reviewed for winter leveling and waterproofing. | Sensors and core components built. Carriage and rail hardware at PDR. Pitch-gate results reported. |

Table 4: Schedule of work breakdown Winter 2027 (Semester 2).

| When | Work | Validation deadline |
|---|---|---|
| Weeks 1–3 | Fabricate rail and carriage. Waterproof electronics. | Electronics sealed enough for spray/splash bench use. |
| Winter week 4 | Engineering Portfolio II (5%). | Waterproofing and fab |

| | | |
|---|---|---|
| | | status reported. |
| Weeks 4–7 | Pitch leveling mechanism designed and built. Integrate gate, carriage, and structure on the bench. | — |
| Winter week 8 | Critical Design Review (15%). | **Carriage mechanism:** stop and gripper swing-clear pass. **Structure:** scaled-force load pass. |
| Weeks 8–10 | Pitch leveling tuned on the rocking rig. End-to-end launches with the scaled-down drone. | **Pitch stabilization:** elevation held in 0–10°. **Launch:** scaled drone leaves undamaged in band. |
| Winter week 11 | Prototype Demo and Technical Defense (25%). | All five validation rows closed (pitch gate, carriage, pitch stabilization, structure, launch). |
| Winter week 12 | Final Report (10%). | — |

## 9. Budget

The budget for the prototype for this launcher is 500 dollars; this includes the price of parts and manufacturing costs. The resources available to us, such as the prototyping shop in the ILC along with the tools and equipment in Mitchell 177, allow us to minimize production costs. A considerable amount of our budget will be allocated to purchasing the Raspberry Pi as well as required sensors and actuators, with the remaining budget being used for the main structure of our launcher. Any additional costs over the 500-dollar budget will be our team's responsibility to cover. The table below shows a rough budget breakdown.

Table 5: Cost Breakdown

| **Item** | **Quantity** | **Price ($)** |
|---|---|---|
| Raspberry Pi 5 | 1 | 199.75 |
| IMU | 2 | 87.36 |
| Stepper Motors | 2 | 110.55 |
| 16mm Spearfishing Bands | 2 | 20.00 |
| 3D Printing Filament | 1 | 18.06 |
| Structure material (acrylic sheets) | 4 | 133.92 |
| Total | | 569.59 |

## 10. Validation

Validation is by subsystem on the scaled prototype before any claim about the boat. Each row has a pass or fail. A hull trial is later, and only if the supervisors grant the boat.

Table 6: validation breakdown

| System | How it is validated | Pass |
|---|---|---|
| Pitch gate | Rocking rig in the few-degree range of the head-sea table. Score the two-cosine forecast against a hold of the last rate. | Fewer false allows than the rate hold during the rail time. |
| Carriage mechanism | Short bench track with rail, latch, brake, and gripper. | Carriage reaches the stop; gripper arms swing clear without striking the body or payload. |
| Pitch stabilization | Same rocking rig. The pitch axis holds a set elevation while the plank moves. | Launch elevation stays inside 0–10°, and the error is logged. |
| Structure | Apply scaled launch forces to the prototype frame and mount. | No permanent bend, crack, or joint slip under the scaled load. |
| Launch | End-to-end release using a scaled-down drone on the prototype. | Aircraft leaves in the elevation band with payload, props, wings, and tail undamaged. |

## 11. Team qualifications

**James Pankiw.** James has a background in software development and in project management for mechanical systems in large industrial bakeries. As a project manager he planned, tracked, and managed ongoing jobs across gas, steam, plumbing, boilers, ventilation, heating, and hot water, including the design of natural gas pipelines that serve bakery ovens and process loads. That work included project economics and profitability, automating administrative tasks, and overseeing installation teams. Overseeing installs built an understanding of mechanical and structural systems, and of fixing unforeseen problems with robust solutions. On this project he brings that planning and software background to the launch gate, sensing, and prediction work.

**Jamison McCulloch-Cop.** Jamison has a strong background in mechanical design with prior work experience in the field. He worked as a mechanical designer for Macleans as an underground mining-vehicle fabricator, and in the field at Cigar Lake mine with Cameco, including wet and flooded conditions. Besides that work experience, he has a strong skill set in control theory, electrical schematics, programming, simulation, and model creation.

**Darcy Ryan.** Darcy has a background working professionally with and on boats, ranging mainly from fast rescue boats to medium-sized work vessels. He has experience working alongside the Coast Guard in Newfoundland while working in the marine rescue and safety training industry. Darcy has also spent time building and maintaining high-pressure systems,

including scuba equipment, compressed-air crane brakes, and compressed breathing-air fill stations.

**Ahrvin Raveendran.** Ahrvin has mechanical design experience for industrial and structural applications, including steel frames and machine components, and experience with CAD modeling and fabrication-ready design in SolidWorks and Fusion 360. He has professionally contributed and gained experience from Nelson Industrial. He has software development experience in a production environment, including development, debugging, and testing from being a Software Systems Intern at the Ministry of Health for the Government of Ontario. He has acquired skills in control systems and embedded programming (PID controllers, Raspberry Pi, Arduino), and has experience with robotics and sensor integration (ROS 2, LiDAR, GPS/IMU) in Linux-based environments from various design teams.

# References

[1] T. Sears, C. Cooper, and J. A. Marshall, “Mapping waves with an uncrewed surface vessel via Gaussian process regression,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2023.

[2] T. M. C. Sears, “Spectral analysis for robotic spatiotemporal mapping,” Ph.D. dissertation, Queen’s University, Kingston, ON, Canada, 2026, ch. 4. [Online]. Available: https://queensu.scholaris.ca/items/ed41fb5e-f0e1-4573-935c-7e94e8f48fed

[3] Insitu Group, Inc., “Methods and apparatuses for launching unmanned aircraft, including methods and apparatuses for launching aircraft with a wedge action,” U.S. Patent 7 128 294, Oct. 31, 2006.

[4] MakeFlyEasy, “Fighter Hand Throw.” [Online]. Available: https://en.makeflyeasy.com/index.php/fighter-hand-throw/

[5] Zodiac Nautic, “PRO 7,” product data sheet. [Online]. Available: https://www.zodiac-nautic.com/